Hacettepe University - Computer Engineering
CMP720 Embedded System Design - Spring 2026
Final Project Topic Proposal

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Project Topic Title

Experimental Latency Characterization and Deterministic Performance Analysis of Real-Time Audio Pipelines on Linux-Based Heterogeneous SoCs

Project Track (choose one)

■ **Implementation Track**

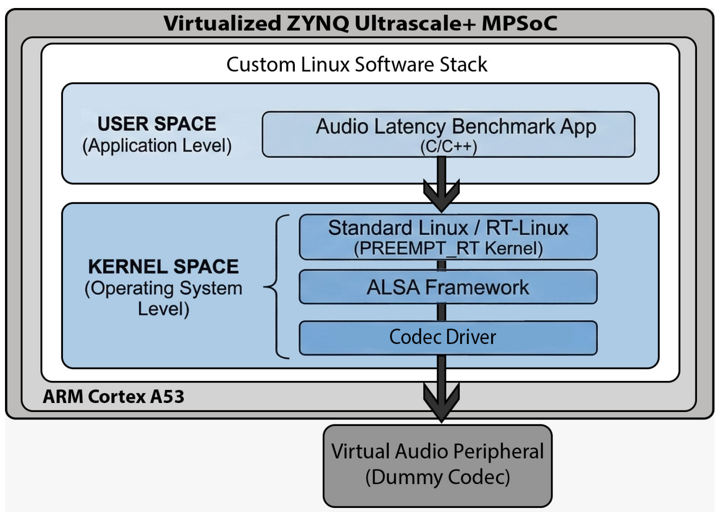
❑ **Optimization Track**

Abstract (300–500 words)

In safety-critical communication systems, the transition from traditional RTOS to Linux-based heterogeneous architectures like Zynq MPSoC introduces significant challenges in maintaining deterministic audio latency. Standard Linux kernels, even with real-time extensions, can exhibit unpredictable jitter and processing delays under high CPU or I/O load, which can lead to audio dropouts and loss of synchronization in mission-critical voice transmissions. This project aims to design and validate a deterministic audio processing pipeline on a virtualized Zynq platform. The methodology follows a dual approach: (i) an experimental characterization inspired by Turchet et al. to benchmark latency and jitter, and (ii) a predictive performance analysis to establish a theoretical latency budget. The analysis focuses on ensuring that end-to-end delays remain within strict deterministic bounds even under simulated hardware contention. The system is built on Renode, simulating a Zynq UltraScale+ MPSoC with four Cortex-A53 cores. The software stack consists of a custom Linux distribution featuring the ALSA (Advanced Linux Sound Architecture) framework. To evaluate timing without hardware dependencies, a virtual soundcard interface will be created.

Initial Plan & Milestones:

- **Environment Setup:** Configuring the Renode Zynq MPSoC platform and booting the custom Linux image with ALSA support.
- **Baseline Benchmarking:** Measuring end-to-end latency in an idle system state using Renode's deterministic Virtual Time logs.
- **Stress Analysis:** Subjecting the pipeline to synthetic CPU and memory bus stress (contention) to measure the impact on jitter and deadline misses.
- **Comparative Validation:** Comparing the performance of standard vs. PREEMPT_RT kernels against the predictive latency budget.



Motivation / Societal Impact (200–500 words)

Motivation: As embedded systems in the defense and telecommunications sectors evolve toward high-performance SoCs, proving the real-time reliability of Linux-based audio pipelines has become a professional necessity. Ensuring that voice data—the most critical data type in emergency communication—is processed with sub-millisecond predictability is vital for the stability of handheld radio devices and networked audio interfaces. This project provides a rigorous simulation-based validation methodology that can be used before moving to physical hardware prototyping.

Societal Impact: Reliable communication is a cornerstone of public safety. In scenarios involving search-and-rescue, disaster management, or law enforcement, a delay or failure in audio transmission can have life-threatening consequences. By establishing a deterministic framework for audio pipelines, this work directly contributes to the development of more resilient and dependable communication tools for first responders and operational teams in high-stakes environments.

Related Work / References

Linux-based embedded audio architecture comparisons are presented in [1]. For timing analysis of event-triggered chains relevant to deterministic pipelines, see [2], while Linux real-time behavior and PREEMPT_RT characteristics are surveyed in [3]. Low-latency audio processing fundamentals are discussed in [4]. The simulation platform used in this project is Renode [5].

References

[1] L. Vignati, S. Zambon, and L. Turchet, "A comparison of real-time linux-based architectures for embedded musical applications," *Journal of the Audio Engineering Society*, vol. 70, no. 1/2, pp. 83–93, 2022.

[2] Y. Tang, N. Guan, X. Jiang, Z. Dong, and W. Yi, "Reaction time analysis of event-triggered processing chains with data refreshing," in *Proceedings of the 60th ACM/IEEE Design Automation Conference (DAC)*, pp. 1–6, 2023.

[3] F. Reghenzani, G. Massari, and W. Fornaciari, "The real-time linux kernel: A survey on preempt_rt," *ACM Computing Surveys*, vol. 52, no. 1, pp. 1–36, 2019.

[4] Y. Wang, *Low Latency Audio Processing*. Phd thesis, Queen Mary University of London, 2012.

[5] Antmicro, "Renode: An open source framework for complex embedded systems simulation," 2024. Official Documentation.

How to Submit

- Open the Google Drive sheet (submission tracker): `https://docs.google.com/spreadsheets/d/17ACM2EVBZ9tZY3XIKveWwRSbt-QfVjS7qkYcTWC3GwU/edit?usp=sharing`
- Fill in your group and project details in an appropriate row (make sure you do not delete/edit other students' entries!).
- Create a repository that you will use to submit **all** project work throughout the semester (GitHub or GitLab).
- Upload/commit your proposal files to the repository (both the `.zip` source files with `.tex` and the compiled `.pdf`).
- Paste the repository link into the corresponding column in the sheet.
- If your repository is private, add the instructor as a collaborator: `selmadilek@gmail.com`.